# Supplementary Material for Separating Training, Evaluation and Constraint-Set Noise in Fairness Uncertainty Quantification

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This supplement collects the full reproduction evidence, the independent verification of the primitive operations, the sample-size sensitivity, and the source ablation. It is self-contained and refers to the main text for definitions. All numbers are produced by the code in the accompanying repository; the files that generate each table are named in the captions.

## 1 Experimental setup and deviations from the original protocol

We reproduce the simulations of Roy and Mohapatra [Roy and Mohapatra, 2026] and extend them with an evaluation-sampling protocol. Table S1 lists the settings actually used. Two deviations were unavoidable and are documented here: (i) the original step size $\eta_k = 0.01\, k^{-0.501}$ is too small to reach $\theta^*$ within the simulated horizons on the synthetic DI generator and on Adult, so we used a calibrated step size ($\eta_0 = 0.1$, and $\eta_0 = 0.03$ for the synthetic DM generator, which has gradient $R^2$-scale); (ii) the barrier coefficient is $\kappa = 0.1$ rather than $10^{-4}$, which keeps the constrained optimum within the evaluated range of trajectories (a $\kappa$-sensitivity analysis is in Table S5). These changes do not affect the qualitative claims: the intervals are valid for the held-out estimand and miss the population estimand exactly as predicted by the asymptotic theory.

Table S1: Settings of the reproduced experiments (results/tables/*_meta_*.json).

| Experiment | $T$ | Reps | $B$ | $\eta_0$ | $\kappa$ | Runtime |
|---|---|---|---|---|---|---|
| Synthetic (DI) | 20000 | 100 | 100 | 0.1 | 0.0001 | 1.1 min |
| Synthetic (DM) | 20000 | 100 | 100 | 0.03 | 0.0001 | 2.1 min |
| Adult (DI) | 50000 | 100 | 200 | 0.1 | 0.1 | 9.4 min |
| ACSIncome (DI) | 50000 | 150 | 100 | 0.1 | 0.1 | 4.3 min |
| COMPAS (DM) | 50000 | 150 | 200 | 0.1 | 0.1 | 13.8 min |

## 2 Complete reproduction results

Table S2 lists every reproduced run of the original simulation protocol; the per-iteration trajectories behind the two W_2 panels are the files in results/raw/.

Table S2: Reproduction of the original simulation protocol (fixed held-out set; results/tables/repro_summary.csv). The interval covers the held-out estimand but not the population gap.

| Experiment | $T$ | Reps | $\phi(\theta^*)$ | Cov. (held-out) | Cov. (population) | CI width | MIS | $W_2$ |
|---|---|---|---|---|---|---|---|---|
| Synthetic (DI) | 20000 | 100 | 0.0835 | 0.89 | 0.43 | 0.0766 | 0.183 | 0.036 |
| Synthetic (DM) | 20000 | 100 | 0.1401 | 0.86 | 1.00 | 0.2084 | 0.248 | 0.030 |
| Adult (DI) | 50000 | 100 | 0.0564 | 1.00 | 0.00 | 0.0219 | 0.028 | 0.027 |
| ACSIncome (DI) | 50000 | 150 | 0.0442 | 1.00 | 0.00 | 0.0135 | 0.021 | 0.034 |
| COMPAS (DM) | 50000 | 150 | 0.0133 | 1.00 | 0.62 | 0.1149 | 0.123 | 0.019 |

# 3 Protocol A: a fresh evaluation set per repetition

In protocol A the constraint set is held fixed and every repetition draws a fresh evaluation set, so that coverage is averaged over the evaluation sampling as well. The baseline interval is the published online multiplier bootstrap; the joint interval resamples training trajectories and the evaluation set. Table S3 gives the complete numbers; the synthetic rows come from joint_bootstrap_v3b_fresh_eval.json and the real rows from real_joint_ci_*.json.

Table S3: Protocol A coverage of the population gap (nominal 0.95) and interval widths. The joint interval is valid in every setting.

| Dataset | $m$ | Cov. (baseline) | Cov. (joint) | Width (baseline) | Width (joint) | Ratio | $s_{\text{train}}$ | $s_{\text{eval}}$ |
|---|---|---|---|---|---|---|---|---|
| Synthetic (DI) | 100 | 0.56 | 0.99 | 0.1069 | 0.2826 | 2.64 | 0.0370 | 0.1002 |
| Synthetic (DI) | 1000 | 0.85 | 0.98 | 0.0773 | 0.1161 | 1.50 | 0.0280 | 0.0318 |
| Synthetic (DI) | 5000 | 0.99 | 1.00 | 0.0743 | 0.0852 | 1.15 | 0.0276 | 0.0142 |
| Adult (DI) | 1000 | 0.30 | 0.98 | 0.0250 | 0.1019 | 4.15 | 0.0067 | 0.0260 |
| ACSIncome (DI) | 1000 | 0.25 | 1.00 | 0.0148 | 0.0821 | 5.72 | 0.0043 | 0.0304 |

# 4 Variance budget

Table S4 separates three sources of variability in a reported gap. The training and evaluation components, estimated from the reproduction interval widths and from the binomial sampling law of the held-out gap, are errors of the reported number around a fixed target. The constraint-set component $s_{D'}$ is the standard deviation of $\Delta(\theta^*(D'))$ over fresh audit designs evaluated on the same pool, so it measures how much the target itself moves; it is a design sensitivity rather than an inferential uncertainty and is not part of the two-source interval used in the coverage experiments. The last two columns compare the average gap over re-drawn designs with the actually reported gap: on Adult and ACSIncome the re-drawn design averages are closer to zero, and on COMPAS the reported gap is more favourable than the design average, changing its sign.

# 5 Barrier-coefficient sensitivity

# 6 Which noise source must be resampled?

Table S6 isolates the contribution of each resampling component (protocol A). The baseline resamples only training trajectories; 'eval only' resamples only the evaluation set; 'train+eval' resamples

Table S4: Three-way variance budget (results/tables/variance_budget_*.json). The last two columns compare the average gap over re-sampled constraint sets with the actually reported gap, quantifying the $D'$ component.

| Dataset | $s_{\text{train}}$ | $s_{\text{eval}}$ | $s_{D'}$ | Train % | Eval % | $D'$ % | $s_{\text{tot}}$ | $\bar{\Delta}_{D'}$ | $\hat{\Delta}_{\text{obs}}$ |
|---|---|---|---|---|---|---|---|---|---|
| Adult (DI) | 0.0056 | 0.0259 | 0.0124 | 3.6 | 78.3 | 18.1 | 0.0293 | -0.0458 | -0.0564 |
| ACSIncome (DI) | 0.0034 | 0.0304 | 0.0245 | 0.8 | 60.3 | 39.0 | 0.0392 | -0.0343 | -0.0442 |
| COMPAS (DM) | 0.0293 | 0.0621 | 0.0753 | 8.3 | 37.1 | 54.6 | 0.1019 | -0.0435 | 0.0133 |

Table S5: Sensitivity to the barrier coefficient on Adult (protocol B, 20 repetitions; results/tables/sensitivity_adult.csv). Coverage is governed by the functional bias, not monotonically by the parameter distance.

| $\kappa$ | $\|\theta^*\|$ | median bias | Cov. (pop.) | CI width | $\phi(\theta^*)$ |
|---|---|---|---|---|---|
| 0.0001 | 10.895 | 10.503 | 0.00 | 0.0399 | 0.1394 |
| 0.01 | 2.204 | 0.808 | 0.95 | 0.0382 | 0.0884 |
| 0.10 | 1.096 | 0.119 | 0.00 | 0.0289 | 0.0570 |
| 1 | 0.315 | 0.025 | 0.60 | 0.0314 | 0.0651 |

both. On the synthetic generators the two sources are comparable in size, so neither single-source interval is valid: with $m = 100$ the training-only and evaluation-only intervals reach 0.56 and 0.98 (DI) and 0.57 and 0.81 (DM), while the joint interval reaches 1.00 and 0.88; at $m = 1000$ the evaluation noise shrinks and the training-only interval remains short (0.85 for DI, 0.78 for DM), whereas the joint interval is at nominal level (0.98 and 0.99). On the real audits the evaluation component dominates, so the training-only interval fails badly while the evaluation-only interval is closer but still short of nominal. Only the combined interval is at or near nominal coverage in every setting.

Table S6: Source ablation, protocol A, nominal 0.95 (results/tables/ablation_*.json).

| Dataset | $m$ | Cov. (train only) | Cov. (eval only) | Cov. (train+eval) | Width (train) | Width (eval) | Width (both) |
|---|---|---|---|---|---|---|---|
| Synthetic (DI) | 1000 | 0.85 | 0.93 | 0.98 | 0.0773 | 0.0938 | 0.1156 |
| Synthetic (DI) | 100 | 0.56 | 0.98 | 1.00 | 0.1069 | 0.2698 | 0.2826 |
| Synthetic (DM) | 1000 | 0.78 | 0.71 | 0.99 | 0.1391 | 0.0734 | 0.1585 |
| Synthetic (DM) | 100 | 0.57 | 0.81 | 0.88 | 0.1513 | 0.2148 | 0.2389 |
| Adult (DI) | 1000 | 0.30 | 0.97 | 0.98 | 0.0250 | 0.0980 | 0.1019 |
| ACSIncome (DI) | 1000 | 0.25 | 1.00 | 1.00 | 0.0148 | 0.0807 | 0.0821 |
| COMPAS (DM) | 400 | 0.62 | 0.97 | 1.00 | 0.0960 | 0.1537 | 0.1734 |

# 7 Independent verification of the primitive operations

A second implementation of every primitive operation (slab projection, multiplier distribution, mean interval score, constrained optimum) was written from scratch and compared with the production code. Table S7 summarises the checks; the scripts are in code/verify/ and the raw JSON files in code/verify/results/.

Table S7: Independent verification summary (notes/verification.md). Check V6 documents that the published step size needs about $10^9$ iterations to reach the constrained optimum on the synthetic generator.

| Check | Content | Status | Key number |
|---|---|---|---|
| V1 | slab projection and KKT conditions | pass | $\max \lvert\tilde{x}^\top \mathrm{II} - \varepsilon\rvert = 8.9 \times 10^{-16}$ |
| V2 | multiplier distribution | pass | $\lvert\mathbb{E}V - 1\rvert = 6.2 \times 10^{-4}$ |
| V3 | mean interval score | pass | max error $1.4 \times 10^{-14}$ |
| V4 | constrained optimum (SLSQP agreement) | pass | KKT residual $3.1 \times 10^{-12}$ |
| V5 | coverage calibration (Gaussian toy model) | pass | clean coverage 0.967 (MC s.e. 0.033) |
| V6 | convergence with the original step size | fail (informative) | $T = 10^6$: relative error 0.414 |
| V7 | elementwise comparison with production code | delayed | covered by V1–V5 at the formula level |

# 8 Empirical convergence rates

Theorem 1 of the main text predicts a $T^{-1/2}$ rate for the averaged iterate. We check this directly on the synthetic DI generator with the constraint set $D'$ held fixed, a fresh evaluation set of size $m = 100$ per replication, $B = 100$ trajectories, 40 replications and the step size $0.1k^{-0.501}$; the truth $\Delta(\theta^*)$ is evaluated on an independent sample of $2 \times 10^6$ observations.

Table S8: Empirical convergence of the averaged iterate (synthetic DI, protocol A, fixed $D'$, 40 replications; raw values in rate_check_synth_di.csv and rate_check_synth_di.json).

| $T$ | $\mathbb{E}\lVert\bar{\theta}_T - \theta^*\rVert$ | $\mathbb{E}\lvert\Delta(\bar{\theta}_T) - \Delta(\theta^*)\rvert$ | $\mathbb{E}[\Delta(\bar{\theta}_T) - \Delta(\theta^*)]$ | $\mathbb{E}\lvert\hat{\Delta}_{100} - \Delta(\theta^*)\rvert$ |
|---|---|---|---|---|
| 1000 | 0.285 | 0.085 | +0.075 | 0.097 |
| 3000 | 0.233 | 0.057 | +0.050 | 0.091 |
| 10000 | 0.167 | 0.047 | +0.038 | 0.082 |
| 30000 | 0.084 | 0.030 | +0.011 | 0.066 |
| 100000 | 0.032 | 0.021 | +0.011 | 0.067 |

The average parameter distance decays at the fitted log–log slope 0.47 and the signed gap bias at 0.46, both consistent with the $T^{-1/2}$ rate (theory $-0.50$); the absolute gap deviations are attenuated because the absolute value of a shrinking random quantity does not retain the power law, and because the $m = 100$ evaluation set contributes a non-vanishing noise floor (last column). For comparison, the Wasserstein distance between the empirical law and the benchmark normal decays like $k^{0.69}$.

# 9 Finite-sample calibration

The asymptotic statements of Theorems 1 and 2 are complemented here by a small finite-sample grid: synthetic DI, protocol A with a fresh evaluation set in every repetition, $B = 100$, $R = 300$, 100 replications, $T \in \{10^3, 3\times10^3, 10^4\}$ and $m \in \{100, 300, 1000\}$; the population gap is evaluated at the constrained optimum on an independent sample of $2 \times 10^6$ observations.

Three features are worth noting. First, the joint interval never falls below the nominal level in this grid (0.98–1.00), while the baseline interval ranges from 0.71 to 0.99; its shortfall grows with

Table S9: Finite-sample calibration of the three intervals (synthetic DI, protocol A, 100 replications; results/tables/ablation_fs_T*_m*.json).

| $T$ | $m$ | Cov. base | Cov. eval | Cov. joint | Width base | Width joint |
|---|---|---|---|---|---|---|
| 1000 | 100 | 0.98 | 0.64 | 1.00 | 0.236 | 0.310 |
| 1000 | 300 | 0.96 | 0.37 | 1.00 | 0.208 | 0.245 |
| 1000 | 1000 | 0.99 | 0.30 | 1.00 | 0.192 | 0.211 |
| 3000 | 100 | 0.87 | 0.83 | 1.00 | 0.215 | 0.330 |
| 3000 | 300 | 0.87 | 0.67 | 0.98 | 0.180 | 0.242 |
| 3000 | 1000 | 0.94 | 0.51 | 0.98 | 0.166 | 0.197 |
| 10000 | 100 | 0.71 | 0.95 | 1.00 | 0.149 | 0.299 |
| 10000 | 300 | 0.79 | 0.88 | 0.99 | 0.130 | 0.204 |
| 10000 | 1000 | 0.92 | 0.86 | 0.99 | 0.116 | 0.146 |

$m$ exactly as the asymptotic analysis predicts. Second, at $T = 10^3$ both intervals over-cover: the baseline absorbs the large training noise into a very wide quantile band that happens to bracket the population gap, whereas the joint interval is conservative—the safe direction for an audit. Third, the width premium of the joint interval is largest exactly where the baseline breaks down (small $m$, large $T$: 0.299 against 0.149 at $T = 10^4$, $m = 100$), which is the price of validity in the regime where the evaluation component dominates.

## 10 Reproduction commands

All experiments run on a single CPU (AMD Ryzen 7 8845H, 16 threads, 27.8 GB RAM, no GPU) under Python 3.13.15 with numpy 2.2.4, pandas 3.0.6, scikit-learn 1.9.1 and scipy 1.18.1. The total wall-clock time of the experiments reported here is about two hours; the finite-sample grid of Table S9 adds about five minutes.

```
python code/data_prep/00_all.py
python code/reproduce_all.py --scale full      # one-command driver for all runs
python code/build_test_tex.py                     # TEST (Springer) LaTeX version
python code/exps/run_synth_di.py
python code/exps/run_synth_dm.py
python code/exps/run_real.py --dataset adult
python code/exps/run_real.py --dataset acsincome
python code/exps/run_real.py --dataset compas
python code/exps/combined_ci_v3_fresh_eval.py
python code/exps/real_joint_ci.py --dataset adult
python code/exps/real_joint_ci.py --dataset acsincome
python code/exps/variance_budget.py
python code/exps/sensitivity_adult.py
python code/exps/rate_check.py --reps 40
python code/exps/ablation_sources.py --dataset synth_di --m 1000
python code/exps/ablation_sources.py --dataset synth_dm --m 1000
python code/exps/ablation_sources.py --dataset adult
```

```
python code/exps/ablation_sources.py --dataset acsincome
python code/exps/ablation_sources.py --dataset compas
python code/exps/ablation_sources.py --dataset synth_di --T 1000 --m 100 \\
    --tag fs_T1000_m100
# finite-sample grid: repeat for all T x m combinations
python code/exps/make_figures.py
python code/exps/make_extension_figures.py
python code/exps/make_sample_size_figure.py
python code/exps/make_ablation_figure.py
python manuscript/check_numbers.py
```

## References

Abhishek Roy and Prasant Mohapatra. Fairness uncertainty quantification: A constrained stochastic optimization perspective. In *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence (UAI)*, volume 337 of *PMLR*, pages 5845–5865, 2026. arXiv:2304.13950.